The pairwise-NHL (PNHL) system takes the form

$$\begin{aligned} \mathrm{d}\boldsymbol{q}_i &= m_i^{-1}\boldsymbol{p}_i \mathrm{d}t, \\ \mathrm{d}\boldsymbol{p}_i &= \boldsymbol{F}_i^{\mathrm{C}}(\boldsymbol{q})\mathrm{d}t + \xi\tilde{\boldsymbol{F}}_i^{\mathrm{D}}\mathrm{d}t, \\ \mathrm{d}\xi &= G(\boldsymbol{q},\boldsymbol{p})\mathrm{d}t - \tilde{\gamma}\xi\mathrm{d}t + \tilde{\sigma}\mathrm{dW}, \end{aligned} \tag{8.111}$$

where all notation is as for PNH and the coefficients $\tilde{\gamma}$ and $\tilde{\sigma}$ are chosen to preserve the target temperature

$$\tilde{\sigma}^2 = 2\tilde{\gamma}\mathrm{k_B T}/\mu, \tag{8.112}$$

and $\mathrm{W} = \mathrm{W}(t)$ is a standard Wiener process.

Splitting methods for the pairwise-NHL (PNHL) system (including those given here) are taken up in [228]. The PNHL system can be divided into five parts such that each piece can be solved exactly, in the sense of distributions:

$$\begin{aligned} \mathrm{d}\begin{bmatrix} \boldsymbol{q}_i \\ \boldsymbol{p}_i \\ \xi \end{bmatrix} = \underbrace{\begin{bmatrix} m_i^{-1}\boldsymbol{p}_i \\ \mathbf{0} \\ 0 \end{bmatrix}\mathrm{d}t}_{\mathrm{A}} + \underbrace{\begin{bmatrix} \mathbf{0} \\ \boldsymbol{F}_i^{\mathrm{C}} \\ 0 \end{bmatrix}\mathrm{d}t}_{\mathrm{B}} + \underbrace{\begin{bmatrix} \mathbf{0} \\ +\xi\tilde{\boldsymbol{F}}_i^{\mathrm{D}} \\ 0 \end{bmatrix}\mathrm{d}t}_{\mathrm{C}} \\ + \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ G \end{bmatrix}\mathrm{d}t}_{\mathrm{D}} + \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ -\tilde{\gamma}\xi\mathrm{d}t + \tilde{\sigma}\mathrm{dW} \end{bmatrix}}_{\mathrm{O}}. \end{aligned} \tag{8.113}$$

Note that the operators corresponding to parts A and B are exactly the same as defined in (8.95) and (8.96), respectively. The operators corresponding to the other terms in the vector field are

$$\begin{aligned} \mathcal{L}_{\mathrm{C}} &= \xi \sum_i \tilde{\boldsymbol{F}}_i^{\mathrm{D}}(\boldsymbol{q},\boldsymbol{p}) \cdot \nabla_{\boldsymbol{p}_i}, \\ \mathcal{L}_{\mathrm{D}} &= G(\boldsymbol{q},\boldsymbol{p})\frac{\partial}{\partial\xi}, \\ \mathcal{L}_{\mathrm{O}} &= -\tilde{\gamma}\xi\frac{\partial}{\partial\xi} + \frac{\tilde{\sigma}^2}{2}\frac{\partial^2}{\partial\xi^2}. \end{aligned}$$

Note that the generator $\mathcal{L}_{\mathrm{O}}$ now only affect the auxiliary variable (as in NHL). The generator for the PNHL system is then defined by

$$\mathcal{L}_{\mathrm{PNHL}} = \mathcal{L}_{\mathrm{A}} + \mathcal{L}_{\mathrm{B}} + \mathcal{L}_{\mathrm{C}} + \mathcal{L}_{\mathrm{D}} + \mathcal{L}_{\mathrm{O}}. \tag{8.114}$$